

1 Schubert et al., 2001 basic equations

This document/section has a different approach to structure the derivation of the ALA: this is a summary of the derivation of the ALA done by Schubert et al., 2001. **red text means assumptions/approximations were made toward the ALA.**

1.1 Mass equation/continuity equation/conservation of mass

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i}(\rho u_i) = \frac{1}{\rho} \frac{D\rho}{Dt} + \frac{\partial u_1}{\partial x_1} = 0 \quad (1)$$

where $\frac{D}{Dt}$ represents the material derivative (??). For incompressible flow, $\frac{D\rho}{Dt} = 0$, so eq 1 reduces to:

$$\frac{\partial u_i}{\partial x_i} = \nabla \cdot \mathbf{u} = 0 \quad (2)$$

For the special case of an incompressible fluid, $\frac{D\rho}{Dt} = 0$, so equation 2 reduces to

$$\frac{\partial u_i}{\partial x_i} = 0 \quad (3)$$

1.2 Momentum equations/conservation of momentum

Starting governing equation (6.4.1):

$$\rho \frac{Du_i}{Dt} = -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} + \rho g_i \quad (4)$$

we assume gravity g to be the only body force. The coriolis force is negligibly small in the Earth's mantle, and the centrifugal force due to the Earth's rotation is included in the gravity term.

In 2D, the momentum equation components are:

$$\rho \left(\frac{\partial u_x}{\partial t} + u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} \right) = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y}, \quad (5)$$

$$\rho \left(\frac{\partial u_y}{\partial t} + u_x \frac{\partial u_y}{\partial x} + u_y \frac{\partial u_y}{\partial y} \right) = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \rho g. \quad (6)$$

Note that the assumption is made that gravity acts in the positive y-direction. Section 6.14?ext

1.3 Navier-Stokes equation

The local strain rate tensor e_{ij} is given by:

$$e_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (7)$$

For a Newtonian, isotropic fluid, the relation between the deviatoric(!) stress tensor and the strain rate tensor is

$$\tau_{ij} = 2\mu e_{ij} + \lambda e_{kk} \delta_{ij} \quad (8)$$

Defining the bulk viscosity k_B :

$$\frac{\tau_{ii}}{3} = e_{ii} \left(\lambda + \frac{2}{3}\mu \right) = k_B e_{ii} \quad (9)$$

The combination of equations 8, 9 and 7 gives

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \left(k_B - \frac{2}{3}\mu \right) \frac{\partial u_k}{\partial x_k} \delta_{ij} \quad (10)$$

and substitution of eq. 10 into eq. 4 gives

$$\rho \frac{Du_i}{Dt} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \left(k_B - \frac{2}{3}\mu \right) \frac{\partial u_k}{\partial x_k} \delta_{ij} \right] + \rho g_i \quad (11)$$

For many fluids, k_B is taken to be zero (**Stokes assumption**), so eq. 10 becomes

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \frac{\partial u_k}{\partial x_k} \delta_{ij} \right) \quad (12)$$

and eq. 11 becomes the Navier-Stokes equation:

$$\rho \frac{Du_i}{Dt} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \frac{\partial u_k}{\partial x_k} \delta_{ij} \right) \right] + \rho g_i \quad (13)$$

for incompressible flow, equation 2 tells us $\frac{\partial u_k}{\partial x_k} = 0$, and together with a constant dynamic viscosity μ so eq. 13 reduces to:

$$\rho \frac{Du_i}{Dt} = -\frac{\partial p}{\partial x_i} + \mu \frac{\partial^2 u_i}{\partial x_j^2} + \rho g_i \quad (14)$$

This equation represents a balance between inertial forces $\left(\frac{Du_i}{Dt} \right)$, pressure forces $\left(\frac{\partial p}{\partial x_i} \right)$, viscous forces $\left(\mu \frac{\partial^2 u_i}{\partial x_j^2} \right)$ and the body force due to gravity (ρg_i)

1.4 Thermal energy equations/conservation of energy

Schubert et al., 2001 presents various different formulations of the energy equation.

The general form of the energy equation (6.9.1) is:

$$\rho T \frac{Ds}{Dt} = \tau_{ij} \frac{\partial u_i}{\partial x_j} + \frac{\partial}{\partial x_j} \left(k \frac{\partial T}{\partial x_j} \right) + \rho H \quad (15)$$

The terms on the right side of 15 are, respectively, volumetric heat production rates due to viscous dissipation (internal friction), thermal conduction, and internal heat generation (radiogenic?). **Where it is assumed that Fourier's law of heat conduction for an isotropic medium is applicable:**

$$q_i = -k \frac{\partial T}{\partial x_i} \quad (16)$$

Where q_i is the heat flux vector. Additionally, the viscous dissipation term can be written as the dissipation function:

$$\Phi = \tau_{ij} \frac{\partial u_i}{\partial x_j} \quad (17)$$

Inserting the definition of our deviatoric stress tensor with $k_B = 0$ (eq. 12), this becomes

$$\Phi = 2\mu \left\{ \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{1}{3} \left(\frac{\partial u_k}{\partial x_k} \right) \delta_{ij} \right\}^2 \quad (18)$$

For an incompressible and Newtonian fluid ($\frac{\partial u_i}{\partial x_i} = 0$), the dissipation function reduces to

$$\Phi = \frac{\mu}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2 \quad (19)$$

With various thermodynamic relationships:

$$\alpha = \frac{1}{v} \left(\frac{\partial v}{\partial T} \right)_p \quad (20)$$

$$c_p = T \left(\frac{\partial s}{\partial T} \right)_p \quad (21)$$

$$\left(\frac{\partial s}{\partial p} \right)_T = - \left(\frac{\partial v}{\partial T} \right)_p \quad (22)$$

It can be shown that

$$\frac{Ds}{Dt} = \frac{c_p}{T} \frac{DT}{Dt} - \frac{\alpha}{\rho} \frac{Dp}{Dt} \quad (23)$$

Plugging 23 into 15 gives

$$\rho c_p \frac{DT}{Dt} - \alpha T \frac{DP}{Dt} = \frac{\partial}{\partial x_i} \left(k \frac{\partial T}{\partial x_i} \right) + \Phi + \rho H \quad (24)$$

$$\rho c_p \left(m dv T t - \frac{\alpha T}{\rho c_p} \frac{DP}{Dt} \right) = \frac{\partial}{\partial x_i} \left(k \frac{\partial T}{\partial x_i} \right) + \Phi + \rho H \quad (25)$$

$$(26)$$

For an incompressible Newtonian fluid (eq. 6.9.13 + 6.9.14):

$$\rho C_v \frac{DT}{Dt} = \frac{\partial}{\partial x_i} + \frac{\mu}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2 + \rho H \quad (27)$$

with $\frac{\mu}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2$ the viscous dissipation function.

In terms of C_p instead of C_v (for a compressible fluid) (eq. 6.9.16):

$$\rho C_p \left(\frac{DT}{Dt} - \left(\frac{\partial T}{\partial p} \right)_s \frac{Dp}{Dt} \right) = \frac{\partial}{\partial x_i} + \frac{\mu}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2 + \rho H \quad (28)$$

with (eq. 6.9.15, 6.9.17):

$$\begin{aligned} \left(\frac{\partial T}{\partial p} \right)_s &= \frac{\alpha T}{\rho C_p} \\ \left(\frac{DT}{Dt} \right)_s &= \left(\frac{\partial T}{\partial p} \right)_s \frac{Dp}{Dt} = \frac{\alpha T}{\rho C_p} \frac{Dp}{Dt} \end{aligned}$$

$$\rho \frac{Du_i}{Dt} = -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} + \rho g_i \quad (29)$$